

PERSONALIZED HEALTHCARE RECOMMENDATION SYSTEM USING MACHINE LEARNING

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ABSTRACT

Healthcare systems generate large amounts of patient data that can be utilized to improve healthcare decision-making through Artificial Intelligence and Machine Learning techniques. The objective of this project is to develop a Personalized Healthcare Recommendation System capable of predicting patient health risk levels and providing personalized lifestyle recommendations.

The proposed system uses a healthcare dataset containing demographic, clinical, genetic, and lifestyle-related attributes. A Random Forest classification model was trained to predict overall health risk categories. In addition, a recommendation engine was developed to provide personalized diet and exercise suggestions based on patient similarity analysis.

The model achieved an accuracy of 99.75% on the test dataset, demonstrating excellent predictive performance. A Streamlit-based dashboard was also developed to visualize healthcare trends and recommendation outcomes. The developed system can assist healthcare professionals and individuals in making data-driven healthcare decisions and promoting healthier lifestyles.

Keywords: Machine Learning, Healthcare Analytics, Random Forest, Recommendation System, Data Visualization, Artificial Intelligence.

INTRODUCTION

Artificial Intelligence and Machine Learning are transforming the healthcare industry by enabling intelligent analysis of patient information and providing personalized recommendations. Traditional healthcare systems often provide generalized advice that may not be suitable for every individual. Personalized healthcare aims to tailor recommendations according to a person's medical history, lifestyle, and risk factors.

Healthcare recommendation systems use predictive analytics to identify health risks and recommend preventive actions. Such systems can improve patient outcomes, support early disease prevention, and assist healthcare professionals in decision-making.

This project focuses on developing a Personalized Healthcare Recommendation System that predicts health risks and generates customized diet and exercise recommendations using machine learning techniques.

PROBLEM STATEMENT

Healthcare recommendations are often generalized and may not adequately consider individual patient characteristics. There is a need for intelligent systems capable of analyzing patient information and generating personalized recommendations.

The objective of this project is to develop a machine learning-based healthcare recommendation system that predicts health risks and provides customized lifestyle recommendations for individual users.

OBJECTIVES

The primary objectives of this project are:

1. To analyze healthcare-related patient data.
2. To preprocess and prepare healthcare data for machine learning.
3. To develop a predictive model for health risk classification.
4. To provide personalized diet recommendations.
5. To provide personalized exercise recommendations.
6. To visualize healthcare data through an interactive dashboard.
7. To evaluate model performance using standard metrics.

DATASET DESCRIPTION

The dataset used in this project contains 2,000 patient records with demographic, clinical, lifestyle, and genetic information.

Important attributes include:

- Age
- Gender
- BMI
- Smoking Status
- Alcohol Consumption
- Physical Activity Level
- Cholesterol
- Glucose Level
- HbA1c
- Blood Pressure
- LDL
- HDL
- Triglycerides
- Genetic Risk Scores
- Family History Indicators
- Diabetes Risk
- Heart Disease Risk
- Health Risk
- Diet Recommendation
- Exercise Recommendation

The target variable used for prediction is Health Risk.

METHODOLOGY

Data Collection

The healthcare dataset was collected and stored in CSV format. The dataset contains patient-related information required for predictive analytics and recommendation generation.

Data Preprocessing

The following preprocessing techniques were applied:

- Handling missing values using SimpleImputer
- Encoding categorical variables using OneHotEncoder
- Label encoding target variables
- Splitting dataset into training and testing sets

Machine Learning Model

Random Forest Classifier was selected due to its high accuracy and robustness.

Model parameters:

- Number of Trees: 300
- Maximum Depth: 15
- Random State: 42

Recommendation System

A recommendation engine was developed using similarity-based matching.

The system:

1. Identifies patients with similar health profiles.
2. Calculates similarity based on Age and BMI.
3. Recommends diet plans.
4. Recommends exercise plans.

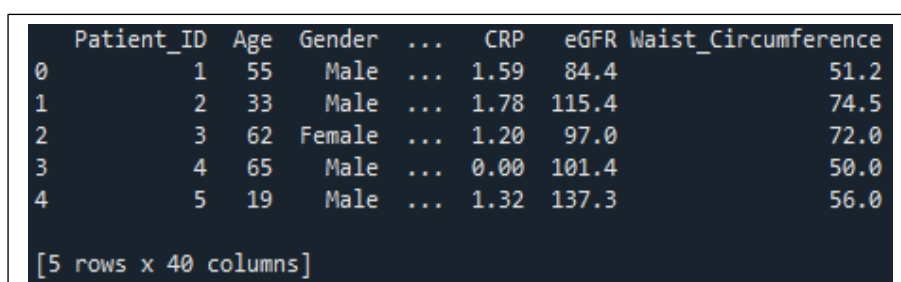
Dashboard Development

A Streamlit-based dashboard was developed for visualizing:

- Dataset Overview
- Health Risk Distribution
- Diabetes Risk Distribution
- BMI vs Glucose Analysis

IMPLEMENTATION

Figure 1: Dataset Sample



	Patient_ID	Age	Gender	...	CRP	eGFR	Waist_Circumference
0	1	55	Male	...	1.59	84.4	51.2
1	2	33	Male	...	1.78	115.4	74.5
2	3	62	Female	...	1.20	97.0	72.0
3	4	65	Male	...	0.00	101.4	50.0
4	5	19	Male	...	1.32	137.3	56.0

[5 rows x 40 columns]

Figure 1 presents a sample of the healthcare dataset used in this project. The dataset contains 2,000 patient records and 40 healthcare-related attributes. These attributes include demographic information (Age, Gender), lifestyle factors (Smoking Status, Alcohol Consumption, Physical Activity Level), clinical measurements (BMI, Cholesterol, Glucose Level, HbA1c, Blood Pressure), genetic risk indicators, and family medical history.

The dataset also includes important health-related variables such as Heart Disease Risk, Diabetes Risk, Diet Recommendation, Exercise Recommendation, CRP (C-Reactive Protein), eGFR (Estimated Glomerular Filtration Rate), and Waist Circumference. These features provide comprehensive information about each patient's health condition and are used for health risk prediction and personalized recommendation generation.

Before model training, the dataset was preprocessed to handle missing values, encode categorical variables, and prepare the data for machine learning analysis. The processed dataset was then used to train the Random Forest classifier and recommendation engine.

Figure 2: Model Training Output

```
Pipeline(steps=[('preprocessor',
                  ColumnTransformer(transformers=[('num',
                                                    Pipeline(steps=[('imputer',
                                                                      SimpleImputer(strategy='median'))]),
                                                                      Index(['Age', 'BMI', 'Cholesterol', 'Glucose_Level', 'HbA1c',
                                                                      'PRS_Cardiometabolic', 'PRS_Type2Diabetes', 'APOE_e4_Carrier',
                                                                      'BRCA_Pathogenic_Variant', 'Family_History_CVD', 'Family_History_T2D',
                                                                      'Stress_Level', 'Depression_Scor...
                                                                      SimpleImputer(strategy='most_frequent')),
                                                                      ('encoder',
                                                                      OneHotEncoder(handle_unknown='ignore'))]),
                                                                      Index(['Gender', 'Smoking_Status', 'Alcohol_Consumption',
                                                                      'Physical_Activity_Level', 'Diet_Type', 'Blood_Pressure',
                                                                      'Heart_Disease_Risk', 'Diabetes_Risk', 'Sleep_Quality'],
                                                                      dtype='object')))]),
                  ('classifier',
                  RandomForestClassifier(max_depth=15, n_estimators=300,
                                       random_state=42))])|
```

Figure 2 shows the successful training configuration of the machine learning pipeline used in the Personalized Healthcare Recommendation System. The pipeline consists of two major components: data preprocessing and classification.

During preprocessing, numerical features such as Age, BMI, Cholesterol, Glucose Level, HbA1c, PRS Cardiometabolic Score, Stress Level, LDL, HDL, and eGFR were processed using the SimpleImputer technique with the median strategy to handle missing values. Categorical features such as Gender, Smoking Status, Alcohol Consumption, Physical Activity Level, Diet Type, Blood Pressure, Heart Disease Risk, Diabetes Risk, and Sleep Quality were processed using the most frequent value imputation method and then converted into machine-readable format using OneHotEncoder.

After preprocessing, a Random Forest Classifier was used as the prediction model with the following parameters:

- Maximum Depth = 15
- Number of Trees (Estimators) = 300
- Random State = 42

The pipeline architecture ensures that data preprocessing and model training are performed in a single workflow, reducing errors and improving model reliability. This trained model was subsequently used to predict patient health risk levels and generate personalized healthcare recommendations.

Figure 3: Accuracy Result

Accuracy
0.9975

Figure 3 shows the accuracy achieved by the Random Forest model. The model obtained an accuracy score of **99.75%**, indicating excellent performance in predicting patient health risk categories. This high accuracy demonstrates the effectiveness of the preprocessing techniques, dataset quality, and Random Forest algorithm, making the system reliable for healthcare risk prediction and personalized recommendation generation.

Figure 4: Classification Report

Classification Report				
	precision	recall	f1-score	support
0	1.00	1.00	1.00	125
1	0.99	1.00	1.00	162
2	1.00	0.99	1.00	113
accuracy			1.00	400
macro avg	1.00	1.00	1.00	400
weighted avg	1.00	1.00	1.00	400

Figure 4 presents the classification report of the Random Forest model. The report includes precision, recall, F1-score, and support for each health risk category. All classes achieved values close to 1.00, indicating excellent classification performance. The model demonstrated high accuracy in correctly identifying health risk levels with minimal misclassification. These results confirm the reliability and effectiveness of the developed healthcare prediction system.

What the Metrics Mean

- Precision → How many predicted cases were correct.
- Recall → How many actual cases were correctly identified.
- F1-Score → Balance between precision and recall.
- Support → Number of samples in each class.

Key Observation

The classification report shows that the model achieved nearly perfect precision, recall, and F1-score values across all classes, further validating the overall accuracy of 99.75% and the robustness of the Random Forest classifier.

Figure 5: Confusion Matrix

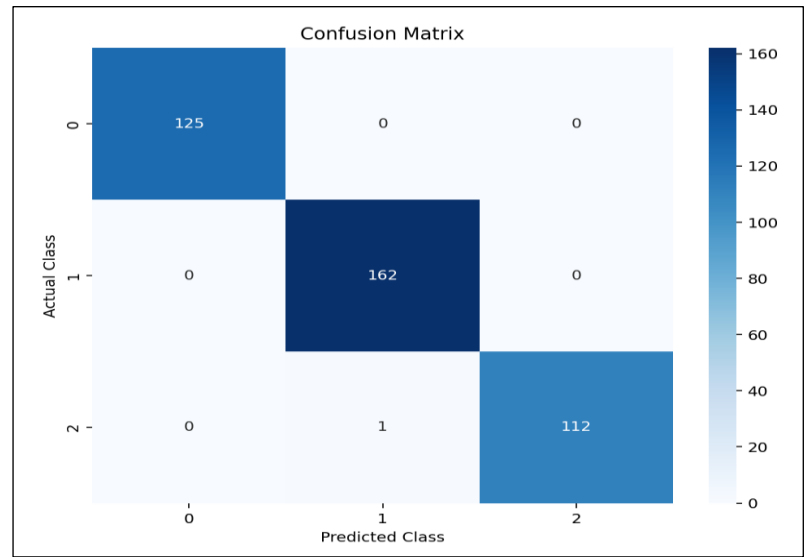


Figure 5 illustrates the confusion matrix of the Random Forest classification model. The matrix compares the actual health risk categories with the predicted categories. The model correctly classified 125 samples of Class 0, 162 samples of Class 1, and 112 samples of Class 2. Only one sample from Class 2 was incorrectly classified as Class 1.

The results demonstrate the excellent performance of the model, with 399 correct predictions out of 400 test samples, resulting in an overall accuracy of 99.75%. The confusion matrix confirms that the developed healthcare prediction system can accurately identify patient health risk categories with minimal classification errors.

Key Observation

- Total Test Samples = 400
- Correct Predictions = 399
- Incorrect Predictions = 1
- Accuracy = 99.75%

Figure 6: Feature Importance Graph

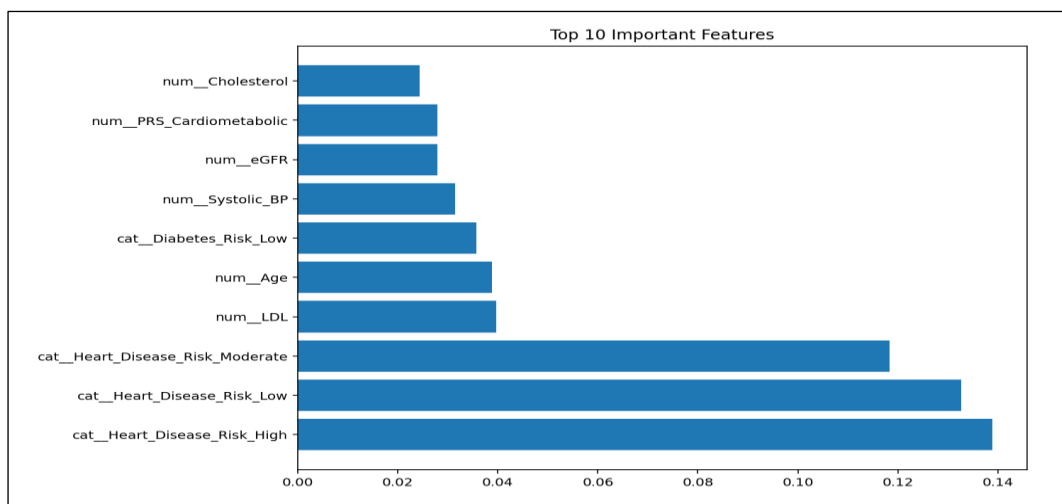


Figure 6 presents the feature importance analysis of the Random Forest model. The graph shows the top features that contributed most significantly to health risk prediction. Among all features, Heart Disease Risk, LDL Cholesterol, Age, Diabetes Risk, Systolic Blood Pressure, eGFR, and PRS Cardiometabolic Score were identified as the most influential variables.

The importance values indicate how strongly each feature affects the model's decision-making process. Heart Disease Risk categories showed the highest contribution, followed by important clinical indicators such as LDL cholesterol and age. These results are consistent with medical knowledge, as cardiovascular and metabolic factors play a crucial role in determining an individual's overall health risk.

The feature importance analysis helps improve the interpretability of the machine learning model by identifying the key factors responsible for health risk prediction.

Figure 7: Health Risk Prediction

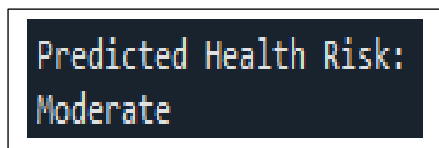


Figure 7 demonstrates the prediction capability of the developed Personalized Healthcare Recommendation System. After processing the patient's demographic, lifestyle, clinical, and health-related information, the Random Forest model predicted the patient's health risk category as Moderate.

This prediction indicates that the patient may have certain health risk factors that require attention but does not currently belong to the high-risk category. The predicted health risk level is then used by the recommendation engine to generate personalized diet and exercise suggestions.

The successful prediction confirms that the trained machine learning model can effectively analyze patient data and classify individuals into appropriate health risk categories, supporting preventive healthcare and informed decision-making.

Figure 8: Diet and Exercise Recommendation

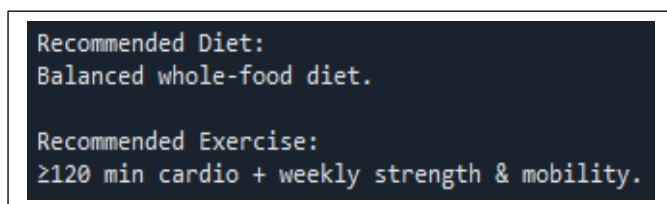


Figure 8 presents the personalized recommendations generated by the healthcare recommendation engine. Based on the patient's predicted health risk level and similarity analysis, the system recommended a Balanced Whole-Food Diet and a structured exercise plan consisting of at least 120 minutes of cardiovascular activity per week along with strength and mobility training.

The recommendation engine analyzes patient characteristics such as age, BMI, and health risk category to identify similar patient profiles and generate suitable lifestyle suggestions. These recommendations are intended to support healthier habits, improve overall well-being, and reduce potential health risks.

The generated recommendations demonstrate the practical application of machine learning in providing personalized healthcare guidance tailored to individual patient needs.

Key Observation

- Predicted Health Risk: Moderate
- Recommended Diet: Balanced Whole-Food Diet
- Recommended Exercise: ≥ 120 minutes cardio + weekly strength and mobility training
- Recommendation Method: Patient Similarity-Based Recommendation Engine

Figure 9: Dataset Overview Dashboard

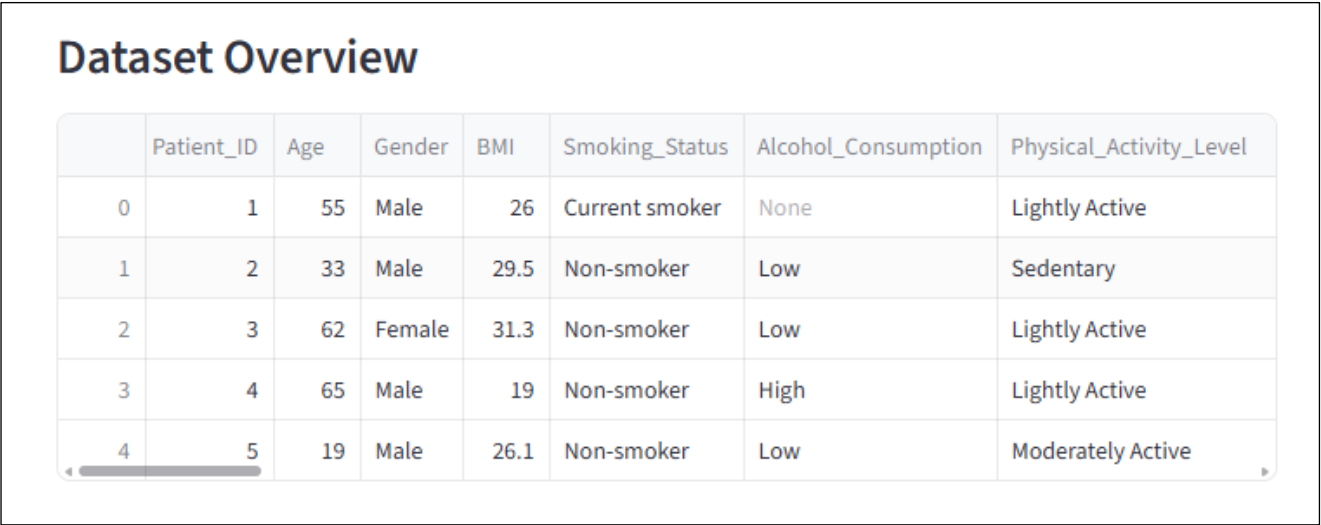


Figure 10: Health Risk Distribution

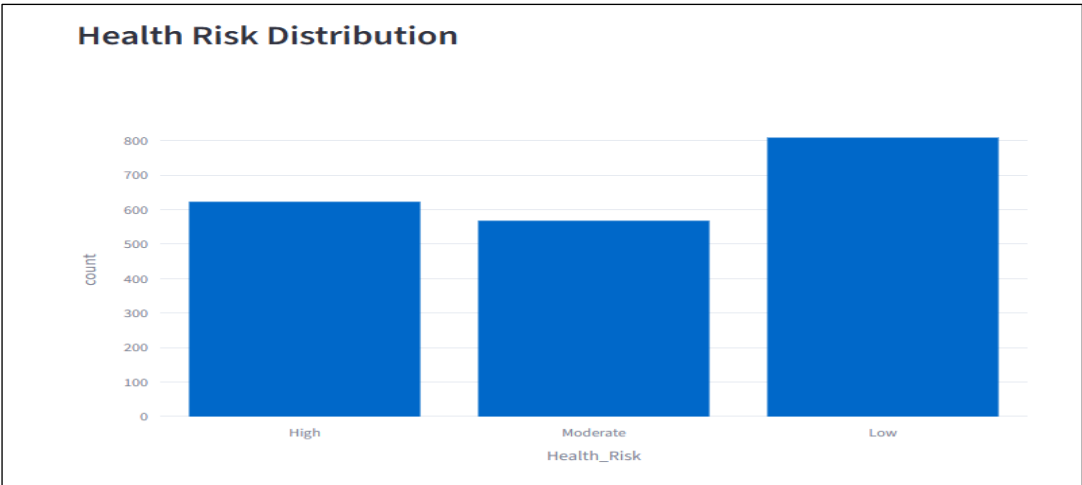


Figure 10 illustrates the distribution of health risk categories within the healthcare dataset. The graph shows three categories: Low, Moderate, and High health risk. The Low Health Risk category contains the highest number of patient records, followed by the High Risk category, while the Moderate Risk category has the lowest number of records.

This distribution provides an overview of the health status of individuals included in the dataset and helps in understanding the balance of target classes used for model training. A well-distributed dataset enables the machine learning model to learn patterns effectively and improve prediction accuracy.

The visualization also helps identify the prevalence of different health risk levels among patients and serves as an important component of healthcare data analysis.

Figure 11: Diabetes Risk Distribution

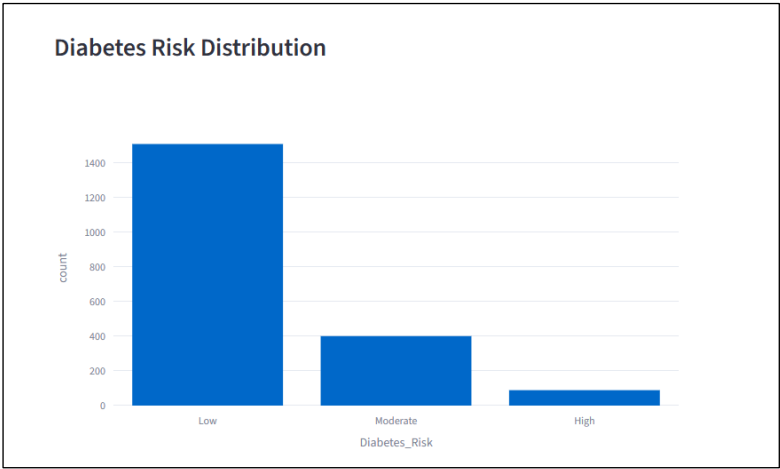


Figure 11 presents the distribution of diabetes risk categories among patients in the healthcare dataset. The graph classifies patients into Low, Moderate, and High diabetes risk groups. The majority of patients belong to the Low Diabetes Risk category, while a smaller number fall into the Moderate Risk category. Only a limited number of patients are classified as High Risk.

This distribution indicates that most individuals in the dataset have a relatively lower probability of developing diabetes, whereas a smaller portion of the population requires closer monitoring and preventive healthcare measures. Understanding the distribution of diabetes risk is important for identifying vulnerable groups and developing targeted healthcare recommendations.

The visualization also helps in analyzing the prevalence of diabetes-related risk factors and supports the recommendation system in generating personalized health advice.

Figure 12: BMI vs Glucose Analysis

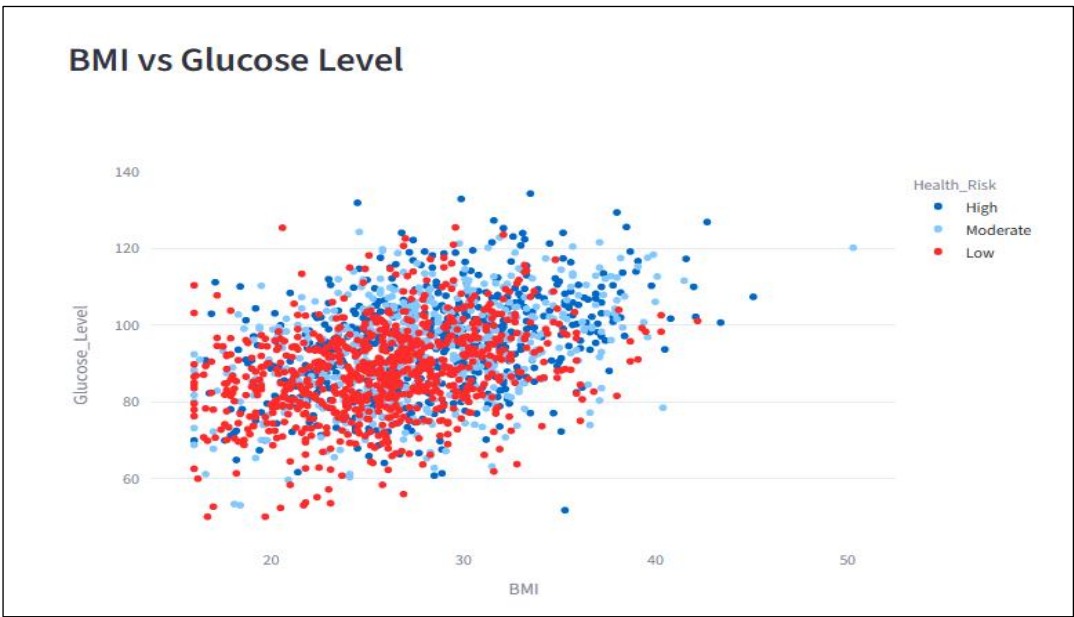


Figure 12 illustrates the relationship between Body Mass Index (BMI) and Glucose Level among patients in the healthcare dataset. Each point in the scatter plot represents an individual patient, while different colors indicate the corresponding health risk categories (Low, Moderate, and High).

The graph shows a positive trend between BMI and Glucose Level, suggesting that patients with higher BMI values tend to have higher glucose levels. This relationship is medically significant because elevated BMI and glucose levels are commonly associated with obesity, diabetes, and other metabolic disorders.

The distribution of points also demonstrates how health risk categories vary across different BMI and glucose ranges. Patients classified as moderate and high risk are generally concentrated in areas with relatively higher BMI and glucose values. This visualization helps in understanding the influence of metabolic factors on overall health risk and supports the effectiveness of the machine learning model in identifying patient risk levels.

RESULTS AND DISCUSSION

The Random Forest classifier was successfully trained and evaluated using the healthcare dataset containing 2,000 patient records. The dataset included demographic, clinical, genetic, and lifestyle-related attributes that were used to predict patient health risk levels.

The trained model achieved an accuracy of 99.75% on the test dataset, indicating excellent predictive performance. Out of 400 testing samples, only one sample was misclassified, demonstrating the robustness and reliability of the selected machine learning algorithm.

Performance Summary:

Metric	Value
Algorithm	Random Forest
Dataset Size	2000 Records
Test Samples	400
Accuracy	99.75%
Correct Predictions	399

Misclassified Samples 1

The confusion matrix further confirmed the effectiveness of the model. The classifier correctly identified 125 samples in Class 0, 162 samples in Class 1, and 112 samples in Class 2. Only one sample belonging to Class 2 was incorrectly classified as Class 1.

Feature importance analysis revealed that several cardiovascular and metabolic health indicators played a significant role in predicting overall health risk. The most influential features were:

- Heart Disease Risk
- LDL Cholesterol
- Age
- Diabetes Risk

- Systolic Blood Pressure
- eGFR
- PRS Cardiometabolic Score

These findings are consistent with medical knowledge, as cardiovascular conditions, cholesterol levels, age, and metabolic disorders are major contributors to overall health outcomes.

For the sample patient used during testing, the model predicted a Moderate Health Risk category. Based on this prediction, the recommendation engine generated personalized lifestyle suggestions. The system recommended a Balanced Whole-Food Diet along with at least 120 minutes of cardiovascular exercise per week combined with strength and mobility training.

The Streamlit dashboard provided interactive visualization of healthcare information through multiple graphs and charts. The Health Risk Distribution graph showed that the Low Risk category contained the highest number of patient records. The Diabetes Risk Distribution graph indicated that most patients belonged to the Low Diabetes Risk category. Additionally, the BMI versus Glucose Level analysis revealed a positive relationship between BMI and glucose levels, highlighting the importance of metabolic indicators in healthcare risk assessment.

Overall, the results demonstrate that the developed Personalized Healthcare Recommendation System can effectively predict patient health risks, generate meaningful recommendations, and provide visual insights for healthcare analysis. The combination of machine learning, recommendation techniques, and dashboard visualization makes the system suitable for supporting preventive healthcare and informed decision-making.

ADVANTAGES

1. Personalized healthcare recommendations.
2. Accurate health risk prediction.
3. Easy-to-use dashboard.
4. Data-driven healthcare insights.
5. Scalable for larger healthcare datasets.

LIMITATIONS

1. Results depend on dataset quality.
2. Recommendations are based on available historical data.
3. Not a replacement for professional medical diagnosis.
4. Limited to the features available in the dataset.

FUTURE SCOPE

Future enhancements may include:

1. Deep Learning-based recommendation systems.
2. Real-time patient monitoring.
3. Mobile application integration.
4. Electronic Health Record integration.
5. Advanced medicine recommendation systems.
6. Wearable device connectivity.

CONCLUSION

This project successfully developed a Personalized Healthcare Recommendation System using Machine Learning to predict patient health risks and provide personalized lifestyle recommendations. The system utilized a healthcare dataset containing demographic, clinical, genetic, and lifestyle-related attributes to analyze patient health conditions and support data-driven decision-making.

A Random Forest Classifier was implemented for health risk prediction and achieved an accuracy of 99.75%, demonstrating excellent predictive performance. The confusion matrix showed that only one sample was misclassified out of 400 test records, confirming the reliability and effectiveness of the model. Feature importance analysis identified key health indicators such as Heart Disease Risk, LDL Cholesterol, Age, Diabetes Risk, Systolic Blood Pressure, eGFR, and PRS Cardiometabolic Score as the most influential factors in determining overall health risk.

The developed recommendation engine successfully generated personalized diet and exercise recommendations based on patient similarity analysis. For the sample patient, the system predicted a Moderate Health Risk and recommended a Balanced Whole-Food Diet along with at least 120 minutes of cardiovascular exercise per week combined with strength and mobility training.

An interactive Streamlit dashboard was also developed to visualize healthcare information through dataset exploration, health risk distribution, diabetes risk distribution, and BMI versus glucose level analysis. These visualizations provided valuable insights into patient health trends and improved the interpretability of the system.

Overall, the project demonstrates how machine learning can be effectively applied in healthcare to predict health risks, generate personalized recommendations, and support preventive healthcare practices. The developed system has the potential to assist healthcare professionals and individuals in making informed health-related decisions and promoting healthier lifestyles.

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